Breno Livio Silva de Almeida

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Computational biology · Biological sequence representation · Machine learning
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Research interests: Biological sequence representation and functional inference in microbial ecology, with a focus on handcrafted descriptors, foundation models and automated machine learning for sequence classification and annotation, complemented by metagenomic metadata curation.

Education

2024 – Present **PhD in Computer Science — University of São Paulo / Leipzig University**
Double-degree programme
Thesis: Decoding the Language of Life: Towards the Limits of Biological Sequence Representation.
Supervisors: André C. P. L. F. de Carvalho and Peter F. Stadler.

2020 – 2024 **BSc in Computer Science, with specialisations in Data Engineering and Data Science — University of São Paulo**
Five-year programme · GPA: 9.4/10 · Rank: 7/116
Research-intensive final year with 600 hours of thesis work.
Thesis: BioDeepFuse: A Hybrid Deep Learning Approach with Handcrafted Features for Non-coding RNA Classification. Paper DOI.
Supervisor: André C. P. L. F. de Carvalho.

Research Experience

07.2025 – 06.2026 **Research Assistant — Helmholtz Centre for Environmental Research (UFZ).**
Helped develop a machine-learning approach to classify data types across millions of runs from the Sequence Read Archive (SRA) within EReMiD. Contributed to metagenomic metadata standardisation in related projects and to the development of teaching materials.

07.2022 – 09.2022 **Research Intern — Big Data Biology Lab, Fudan University.**
Developed a rarefaction workflow for nearly one billion unique small open reading frames (smORFs) across 63,402 samples. Contributed to *A catalog of small proteins from the global microbiome* (*Nature Communications*, 2024). Lab blog post · Paper.

Research Funding and Fellowships

Awarded approximately **€110,181** in competitive fellowships, research scholarships, and mobility/travel grants across the award periods.

2025 **Google PhD Fellowship in Health Research** — USD 30,000.
Selected among 15 fellows in Latin America in 2025, including eight from Brazil. Recipients · USP award report.

2025 **ISME Scholar Mobility Fund** — EUR 2,300.
Awarded by the International Society for Microbial Ecology for a research visit to UFZ, Leipzig, in July 2026. Programme · USP visit report.

2024 **FAPESP Doctoral Scholarship** — BRL 453,168.
Decoding the language of life: Towards protein classification and design through large language models. São Paulo Research Foundation (FAPESP); competitive, peer-reviewed; grant #2024/10958-1. FAPESP grant record.

Indicative total at fixed reporting rates: BRL 1 = EUR 0.17; USD 1 = EUR 0.85. These are indicative conversions, not current market rates. Total excludes cash prizes, estimated training values, and consortium budgets; it represents award values, not payments received.

Honours and Awards

2026 **IRCAI Global Top 100 — Outstanding (2025 edition).**
Team recognition for BioAutoML from the International Research Centre on Artificial Intelligence (IRCAI), under the auspices of UNESCO. Official recognition.

2026 **HIDA-sponsored Carpentries Instructor Training.**
Selected for one of 15 annually sponsored places for Helmholtz researchers; estimated training value: USD 1,600. HIDA programme and selection details.

2024 **J. F. Marar Artificial Intelligence Prize for Undergraduate Research.**
Winner for *BioDeepFuse: Empowering Researchers in Life Sciences with Deep Learning*; cash prize: **BRL 10,500**. USP award announcement.

2024 **Prototypes for Humanity.**
BioAutoML selected among 100 projects from nearly 3,000 submissions worldwide. Selected project.

Publications

126	77	4	3
Citations Google Scholar	Citations Scopus	h-index Google Scholar	i10-index Google Scholar

Google Scholar as of 2026-10-19 · Scopus as of 2026-10-19.

† Equal contribution.

Manuscripts Under Review

- **de Almeida, B. L.**, Bonidia, R. P., Bole, M., Avila-Santos, A. P., Stadler, P. F., Nunes da Rocha, U., & de Carvalho, A. C. P. L. F. (2026). BioAutoML-FAST: an automated machine-learning platform for reusable and benchmarked biological sequence models. *bioRxiv*. **DOI:** 10.64898/2026.04.18.719383. **Under revision at *Bioinformatics* following peer review (major revision).**

Published Journal Articles

- Duan, Y., Santos-Júnior, C. D., Schmidt, T. S., Fullam, A., **de Almeida, B. L.**, Zhu, C., … & Coelho, L. P. (2024). A catalog of small proteins from the global microbiome. Nature Communications, 15(1), 7563. **DOI:** 10.1038/s41467-024-51894-6.
- **de Almeida, B. L.**†, Avila Santos, A. P.†, Bonidia, R. P., Stadler, P. F., Stefanic, P., Mandic-Mulec, I., … & de Carvalho, A. C. (2024). BioDeepfuse: a hybrid deep learning approach with integrated feature extraction techniques for enhanced non-coding RNA classification. RNA Biology, 21(1), 410-421. **DOI:** 10.1080/15476286.2024.2329451.
- Bonidia, R. P., Avila Santos, A. P., **de Almeida, B. L.**, Stadler, P. F., Nunes da Rocha, U., Sanches, D. S., & De Carvalho, A. C. (2022). Information theory for biological sequence classification: A novel feature extraction technique based on Tsallis entropy. Entropy, 24(10), 1398. **DOI:** 10.3390/e24101398.
- Bonidia, R. P., Santos, A. P. A., **de Almeida, B. L.**, Stadler, P. F., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. (2022). BioAutoML: automated feature engineering and metalearning to predict noncoding RNAs in bacteria. Briefings in Bioinformatics, 23(4), bbac218. **DOI:** 10.1093/bib/bbac218.

Accepted Journal Articles

- **de Almeida, B. L. S.**†, Bonidia, R.†, Florentino, B.†, Barreto, M. C. A.†, Constantino, J. L. R.†, Tavares da Silva, D.†, Zwicker, G. G., Rocha, U., Goedert, G., Struchiner, C., & de Carvalho, A. AutoAI Hub: From data to society, democratizing machine learning for everyone. Oxford Open Digital Health. **Accepted for publication**.
- Tavares, D., Barreto, M. C. A., **de Almeida, B. L. S.**, Florentino, B. R., Tetzner, F., Goedert, G., Struchiner, C., Bonidia, R., & de Carvalho, A. Dominique: An AI-powered fact-checking chatbot for democratizing access to reliable information. Oxford Open Digital Health. **Accepted for publication**.

Conference Proceedings

- **de Almeida, B. L. S.**†, Queiroz, A. P.†, Santos, A. P. A.†, Bonidia, R. P., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. P. D. L. F. (2021, November). Feature importance analysis of non-coding DNA/RNA sequences based on machine learning approaches. In Brazilian Symposium on Bioinformatics (pp. 81-92). Cham: Springer International Publishing. **DOI:** 10.1007/978-3-030-91814-9_8.

Research Software and Resources

BioAutoML-FAST — Automated machine learning for biological sequences. Lead developer; expanded sequence types, prediction tasks, and evaluation within the BioAutoML framework. Code · Web application · Preprint.
BioAutoML — Automated feature engineering and model selection. Contributing developer. Code · Paper.
BioDeepFuse — Hybrid deep learning for non-coding RNA classification. Co-first author of the methods paper. Combines convolutional or bidirectional LSTM networks with handcrafted sequence features to capture complementary sequence patterns. Code · Paper.
MAG2ENA — Metagenome-assembled genome submission to the European Nucleotide Archive. Lead developer; developed within NFDI4Microbiota. Code · Web application.

Selected Presentations

2026 **ISME20, Auckland** — oral presentation. Protein language models for protein-family annotation in environmental metagenomes.
2026 **VAAM, Berlin** — oral presentation. Protein language models annotating 453,261 additional orthogroups across microbial evolutionary diversity: 93 bacterial and 10 archaeal phyla.
2025 **Helmholtz AI, Messe Karlsruhe** — oral presentation. Protein language models for domain and family annotation of 183,044 novel orthogroups.

Teaching and Mentoring

2025 – 2026 **Instructor — UFZ Summer School on Trends in Multi-Omics Data Analysis for Microbial Ecology and Biotechnology**, Leipzig.
Taught Unix/Bash/Git, machine learning, and sequence annotation to approximately 30 participants across two editions, delivering approximately 20 hours of instruction. Teaching materials (GitHub).
09.2025 **Instructor — NFDI4Microbiota workshop, 13th International Symposium on Anaerobic Microbiology (ISAM)**, Leipzig.
Taught submission of research data and metadata to public repositories to approximately 20 students.
06.2025 **Co-organiser and instructor — Metagenomics workshop, 17th Symposium on Bacterial Genetics and Ecology (BAGECO)**, Graz.
Taught experimental design, computer experiments, and end-to-end sequence analysis with BioAutoML to approximately 20 students. Workshop programme.
05.2021 – 11.2022 **Teaching Assistant — University of São Paulo.**
Eight hours/week across Object-Oriented Analysis and Design, Statistics, Fundamentals of Data Science, and Computational Visualization, serving approximately 150 students in total.

Academic Service

Journal Peer Review — *Computational and Structural Biotechnology Journal. Journal of Thermal Biology. Briefings in Bioinformatics.*
Member of six undergraduate thesis examination committees.
2018 – 2021 **Co-founder and coordinator — efestus**, University of São Paulo.
Student organisation addressing computing, data science, and technology ethics for social good. Funded participation in the TechShift Summit, Chapel Hill, USA.

Technical Skills

Programming	Python, R, Bash, Nix, SQL, Java, C, C++.
Modelling and data	Scikit-learn, TensorFlow, pandas, Polars, NumPy, SciPy, NetworkX.
Sequence analysis	Biopython, BLAST, Prodigal, OrthoFinder, Anvi'o, HMMER, Infernal, DIAMOND, MMSeqs2.
Research tools	Git, Linux, Jupyter, R Markdown, LaTeX, Typst, Matplotlib, Plotly, Streamlit.
Languages	Portuguese (native), English (C1), German (A1.2).